

CCGL9065

Our Response to Climate Change:

Hong Kong 2100

Final Portfolio Submission - Essay

“Deadly Ring Tightening Humanity’s Future”:

Increase of Space Debris continuously nibbling the road to the Universe

Rockets and spaceships sound like one of the coolest things for most of us in our childhood, which symbolize the top point of human technology, and the desire for exploration. However, when we shift our gaze from these small boats carrying humanity’s hopes as they pierce through the clouds, back to the geostationary orbit. Tons of debris are forming a shocking but also dangerous “steel star belt” around our planet.

The amount of larger debris with size **larger than 10 cm** reach up to 29,000 pieces, and smaller ones with size larger than 1cm and 1mm reaches **670,000** and **190,000,000** respectively.¹ According to the ESA Space Environment Report 2025, there were about 11 to 12 accidental fragmentations per year, and the 11 break-up events in 2024 generated **over 2600 new pieces** of trackable debris.² This number will become even larger in future, since the amount of fragments increase, the possibility of collision also increase, then creating even more after collision, spreading the cloud of death further. This cloud may even become a large obstacle in humanity’s future development in space: satellites between Low-Earth Orbit and Geostationary Orbit is facing an extremely high probability of being shot down by space debris, which may have a widespread influence, including defense sector, airline industry, marine industry, railway industry, roadways, GNSS timing, agriculture.

Solving this systems-level crisis requires moving beyond voluntary guidelines toward mandatory orbital "traffic laws" and active extraction. The first step is the enforcement of the "Five-Year Rule." In late 2024, the FCC shifted from a 25-year recommendation to a strict five-year deadline for satellite disposal after mission

¹ Mariappan, A., & Crassidis, J. L. (2023). Kessler’s syndrome: a challenge to humanity. *Frontiers in Space Technologies*, 4. <https://doi.org/10.3389/frspt.2023.1309940>

² ESA Space Environment Report 2025. (n.d.).
https://www.esa.int/Space_Safety/Space_Debris/ESA_Space_Environment_Report_2025

completion.³ This creates a "leave no trace" policy for new launches, preventing the next generation of debris. However, regulation only stops the bleeding; it doesn't remove the 2,600 new fragments generated just last year.

The second, more critical step is Active Debris Removal (ADR). We must treat the orbit like a national park that requires maintenance. Missions like **ClearSpace-1** and Astroscale's **ELSA-M** have moved from theory to deployment in 2025-2026⁴, using robotic arms and magnetic docking to capture high-risk objects. By targeting the "statistical triggers"—the large, defunct boosters that act as collision bombs—we can prevent the Kessler Syndrome before it becomes self-sustaining. Furthermore, shifting to a "Circular Space Economy" where companies are incentivized to refuel and repair satellites rather than abandoning them turns orbital space from a graveyard into a sustainable industrial zone.

The payoff of this smart management is the survival of our digital infrastructure. Protecting the LEO and Geostationary ecosystems secures a global space economy projected to hit \$1.8 trillion by 2035⁵. If we fail, the "cloud of death" will eventually blind our weather satellites and collapse the GNSS timing that synchronizes our global power grids and financial markets. We aren't just saving a view; we are defending the nervous system of modern civilization.

Individuals can accelerate this shift by demanding transparency from the private sector. Support the **Zero Debris Charter** by favoring telecommunication providers that commit to debris-neutral launches. Public pressure creates the market incentive for companies to invest in expensive cleanup technology. When the cost of polluting space exceeds the cost of cleaning it, the industry will pivot.

As the orbit dies, our connectivity and safety die with it. But by choosing to clean our skies, we ensure that the top point of human technology remains a gateway

³ Federal Communications Commission (2024). Second Report and Order: Five-Year Rule for Post-Mission Disposal of Satellites.

⁴ Astroscale (2025). ELSA-M Mission: Commercial Debris Removal Operations.

⁵ World Economic Forum (2026). The \$1.8 Trillion Opportunity: Protecting Orbital Assets.

to exploration rather than a self-imposed prison. The stars are still within reach, provided we clear the path today.

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